INTEGRALITY OVER AN IDEAL IS EQUIVALENT TO INTEGRALITY OF ST OVER A REES ALGEBRA

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Allowed standard prerequisites (editorial).

- The Rees algebra A of an ideal as a subring of R[t]. Defined in the statement at lines 8586-8588 and used throughout the counted proof. Supplies the ring over which the element st is tested for integrality and in which the degree decomposition takes place.
- Degree decomposition in a graded subring. Used at proof lines 8598-8602. Lets the coefficients of a monic relation be replaced by their single relevant homogeneous parts.

Lemma 0.1. *Let $\varphi : R \to S$ be a ring map. Let $I \subset R$ be an ideal. Let $A = \sum I^n t^n \subset R[t]$ be the subring of the polynomial ring generated by $R \oplus It \subset R[t]$. An element $s \in S$ is integral over I if and only if the element $st \in S[t]$ is integral over A.*

Proof. Suppose st is integral over A. Let $P = x^d + \sum_{j<d} a_j x^j$ be a monic polynomial with coefficients in A such that $P^\varphi(st) = 0$. Let $a'_j \in A$ be the degree $d - j$ part of a_j, in other words $a'_j = a''_j t^{d-j}$ with $a''_j \in I^{d-j}$. For degree reasons we still have $(st)^d + \sum_{j<d} \varphi(a''_j) t^{d-j} (st)^j = 0$. Hence $s^d + \sum_{j<d} \varphi(a''_j) s^j = 0$ and we see that s is integral over I.

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Suppose that s is integral over I. Say $P = x^d + \sum_{j<d} a_j x^j$ with $a_j \in I^{d-j}$. Then we immediately find a polynomial $Q = x^d + \sum_{j<d}(a_j t^{d-j})x^j$ with coefficients in A which proves that st is integral over A. $\square$